Day 4 - Weekly Check-In

# Open Agent Hackathon




POWERED BY:

GenAI.works METERLESS

# MEET THE INSTRUCTOR



Lead Instructor

**Samuel Cummings**

Data Scientist & Director of Education @ GenAI.Works




ST. LOUIS BUSINESS JOURNAL

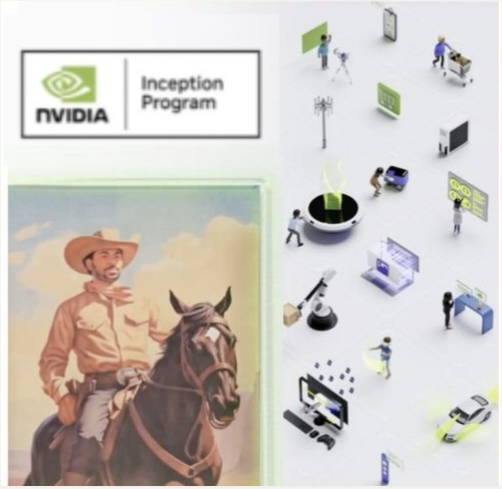
Startups to Watch 2023: Dataplant aims to help other startups grow revenue

Miami Herald

Part of the McClatchy Media Network

News Sports Business Politics Opinion Food & Drink

**Generative AI Black tech founder wins investment by traveling home to St. Louis**



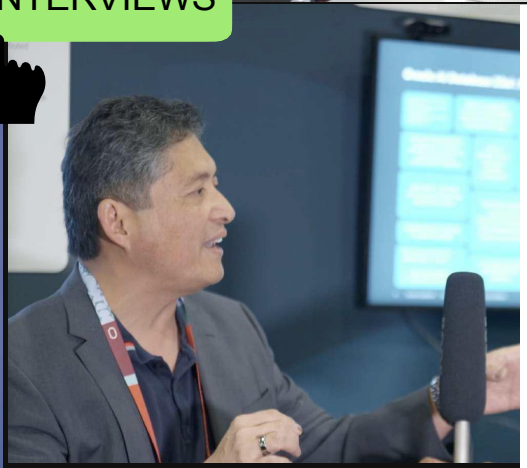






OVER 60+ INTERVIEWS





Meet The Instructor

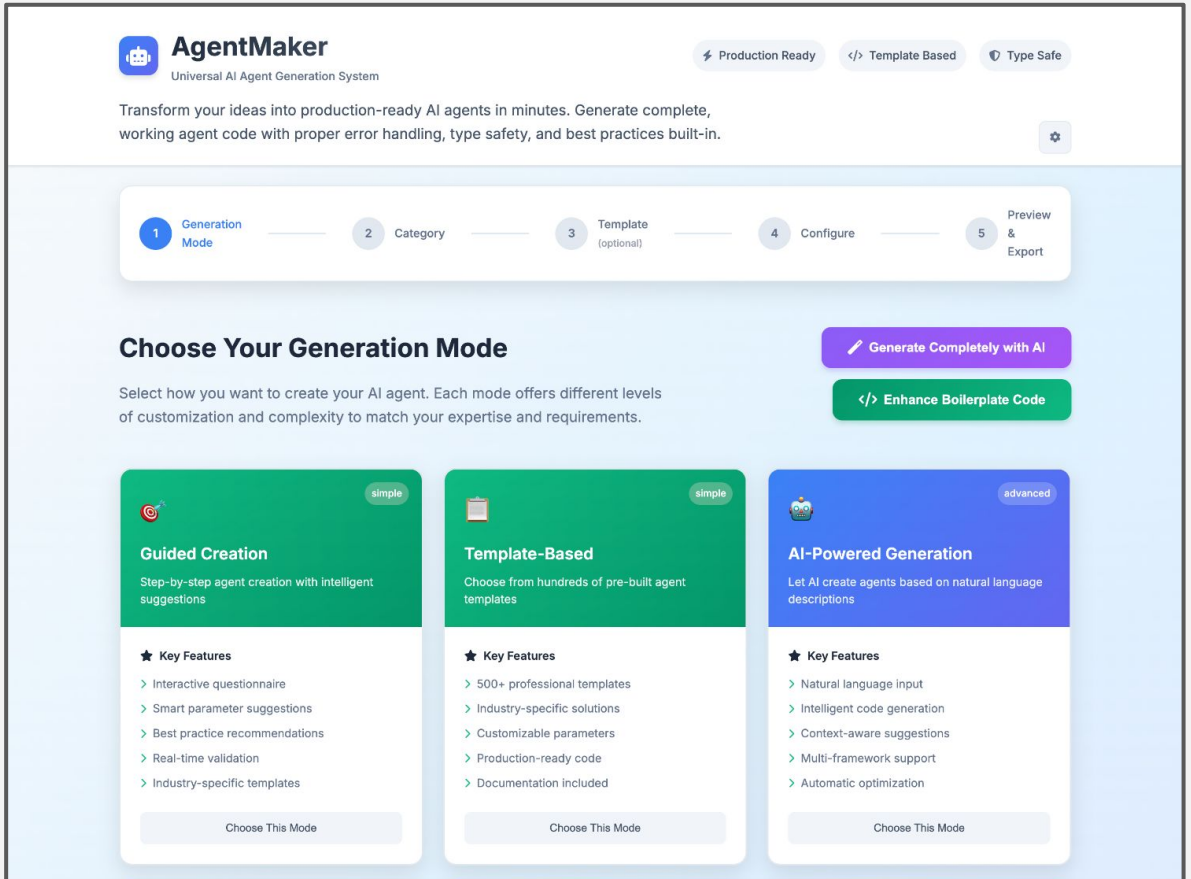


Lead Instructor

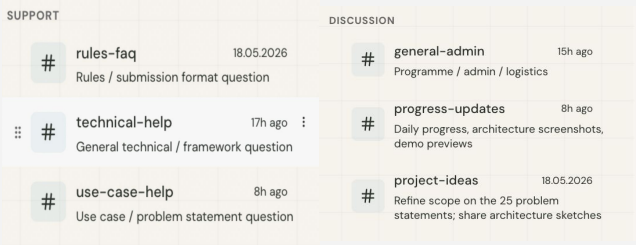
Samuel Cummings

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# LET'S TALK ABOUT AGENT CREATION




- Is it a hard problem or a complex one?
- What are the elements of an agent's jobs?
- How do we compare agents universally?




YOU WILL NEED TO POST TO CHANNELS

# COMPARE AGENTS ON JOBS TO BE DONE



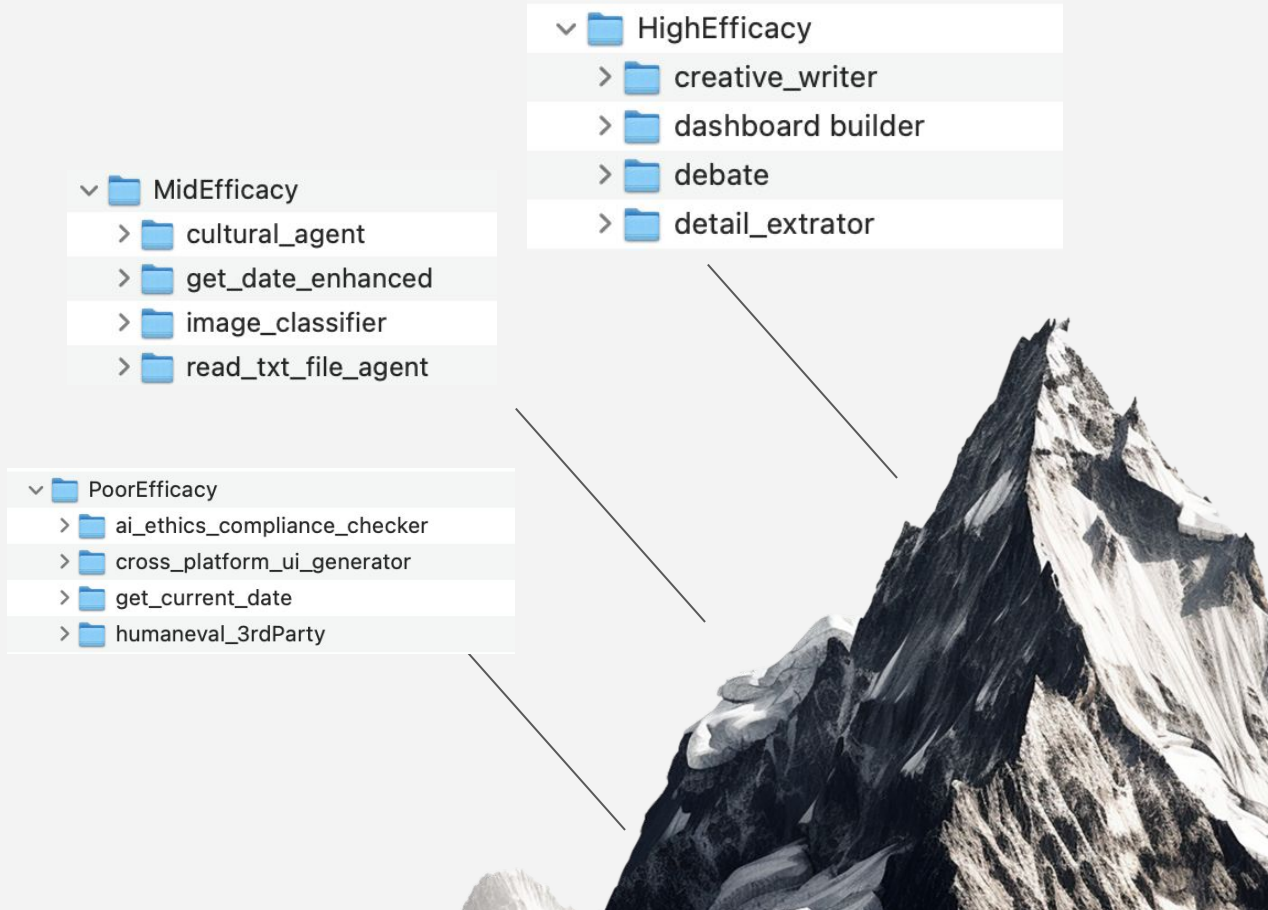
1. **Workflow Structuring:**
    a. Ask: “What’s the best workflow for managing team tasks?”

2. **Feature Prioritization:**
    a. Ask: “Rank features for MVP by impact and complexity.”

3. **Wireframe Creation:**
    a. Ask: “Describe a homepage wireframe for our app.”

# CLIMBING AGENTIC ENTROPY MOUNTAIN

- Entropy represents the complexity of the code for your agent or its job to be done
- Agents are relatively low entropy individually but combine
- You can measure entropy in agentic systems in two ways
  - Complexity
  - Efficacy



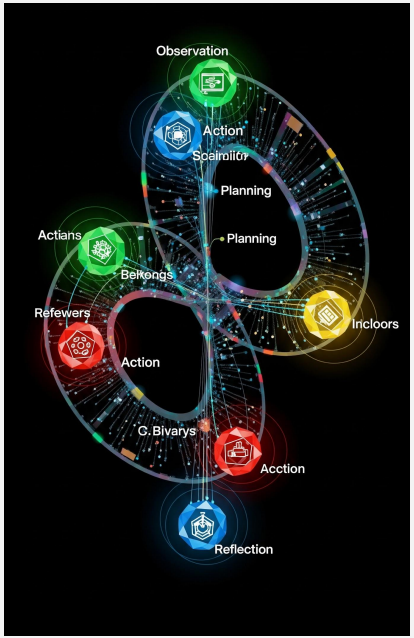
# THE PROBLEM OF VALIDATING THE EFFICACY




- How do you can you tell how complex an application is?
- How do I verify the levels of complexity:
  - Visual
  - Functional
  - Outputs
- Visual & Output is easy to see but I might not have full visibility
- Functional is what you can do or it does under the hood but there is no direct way to confirm.

# Time to work + Open Q&A

# SPECIFYING AGENTIC AI LOOPS TOGETHER




**Agent Loop Spec (fill in 8 lines):**

1. **Goal:** _____
2. **Plan (≤5 steps):** [ ] [ ] [ ] [ ] [ ]
3. **Tools (name → inputs → output → fail mode):** _____
4. **Memory to keep:** scratchpad / decisions / best template
5. **Rubric (0–3 ×3):** accuracy • appropriateness • actionability
6. **Stop conditions:** if score <7/9 → ask human
7. **Run log:** step • decision • result • next
8. **Guardrails:** cap steps; cap tool calls; hard stops; always log *why*.

**Take a sec to document the following for yourself and after the call post yours in the team chat created under your team name**

# Building Better Agents

# FIVE AREAS TO BOOST AGENTIC EFFICACY

Agents are powered by these core capabilities:

**Autonomy**
How far the agent can run independently without human intervention.

**Reasoning & Planning**
Turning high-level goals into clear, checkable execution steps.

**Memory**
Scratchpad → Episodic log → Reusable knowledge notes.

**Tool Use**
Using APIs and apps as verbs with structured inputs and outputs.

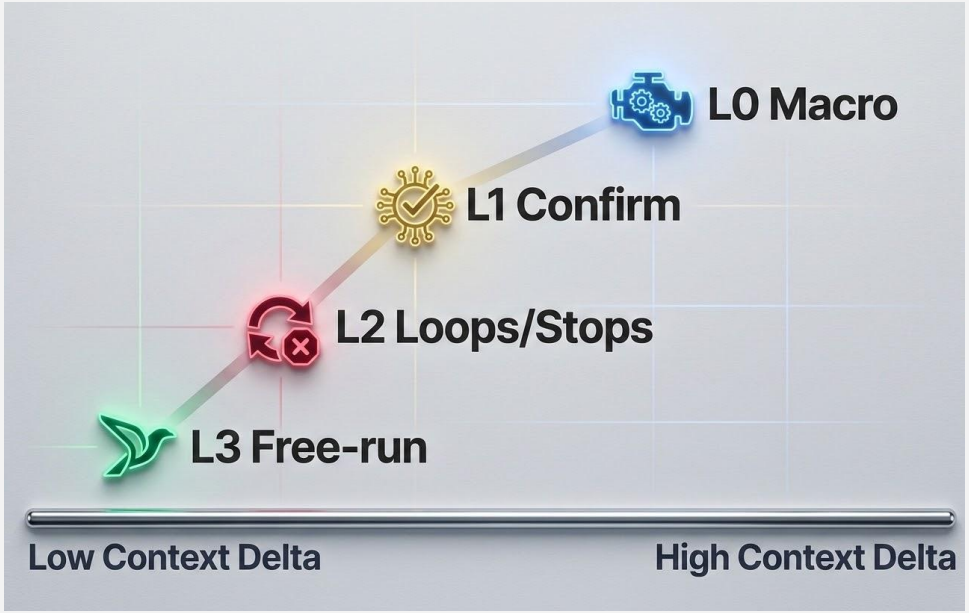
**Governance**
Defining the essential rules and safety guidelines for the agent.



# HOW TO PICK THE RIGHT LEVEL OF AUTONOMY

There are 4 levels of autonomy

- **L0 Macro:** one-shot; human executes steps
- **L1 Confirm:** agent proposes, you approve
- **L2 Loop/Stops:** agent iterates, must pass checks
- **L3 Free-run:** rare; only for highly repeatable tasks

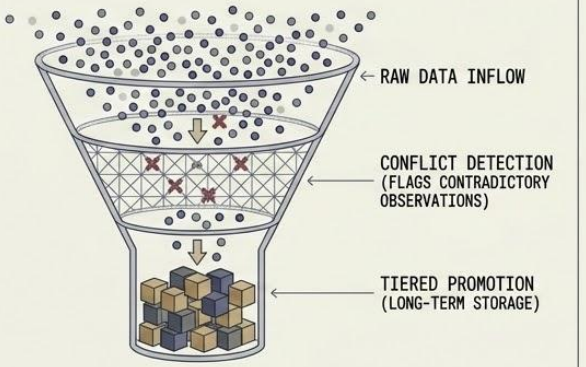



Re-evaluate your current Agentic Loop and list yours in the project ideas channel

# MASTERING AGENTIC PRIMITIVES

## MEMORY

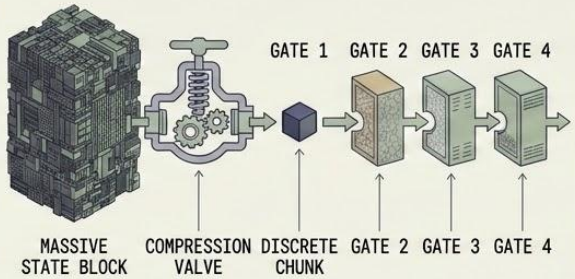
### H-MEM (HIERARCHICAL MEMORY)




- Biological 3-tier memory. Ends blank-slate sessions.
- Dreaming: AI-powered cross-session pattern synthesis during sleep cycles.

## REASONING

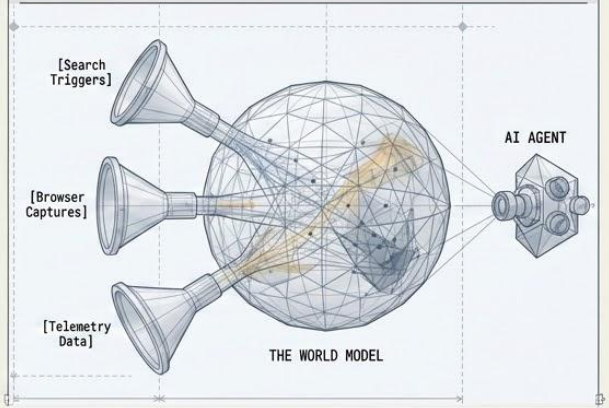
### MARKOVIAN ENGINE




- Bounded-growth context scaling.
- Decomposes massive tasks. Passes only compressed essential state forward. Compute cost stays flat.

## TOOL USE

### WORLD MODEL




- Store all of the entities, relationships
- One shared world between agents and processes that evolves and adapts

Download each from the Resources Tab in the Agentathon App, Find More On Github

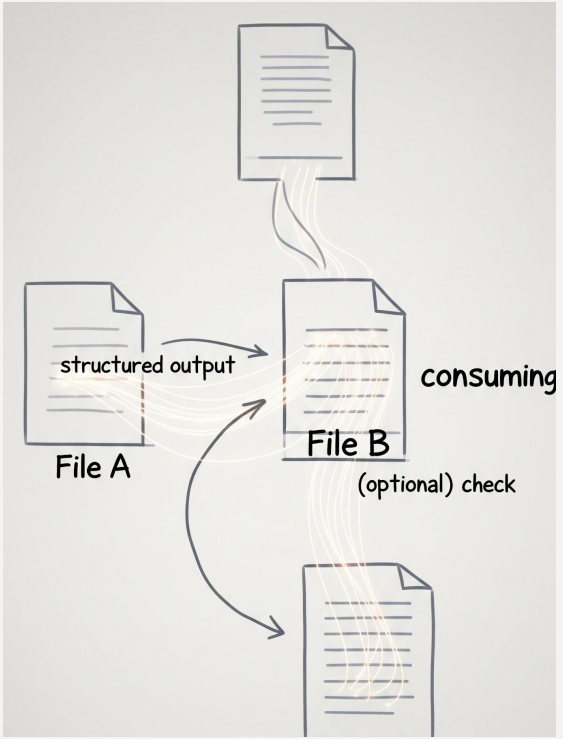
# GOVERNANCE POWERED BY AGENTIC PAIRINGS

You don't need an "agent tool" to govern work agentically. Wrap your workflows in agentic pairings.

Full autonomy is fragile governance wise.

Human-in-the-loop + light autonomy wins for most knowledge work governance use cases.

- Use Questions Upfront
- Finding the Right Verification Protocols
- Combine and Stack Agentic Pairings



# GOVERNANCE POWERED BY AGENTIC PAIRINGS

**Planner + Doer**

- `planner.md` → turns a **GOAL** into ≤5 steps with inputs/outputs & tool hints
- `doer.md` → executes **step-by-step,** logs results, stops on rubric fail

**Proposer + Critic**

- `propose.md` → drafts full (copy/brief) & (plan/decision)
- `critic.md` → scores against a rubric; if <7/9, returns **one fix** then revise once

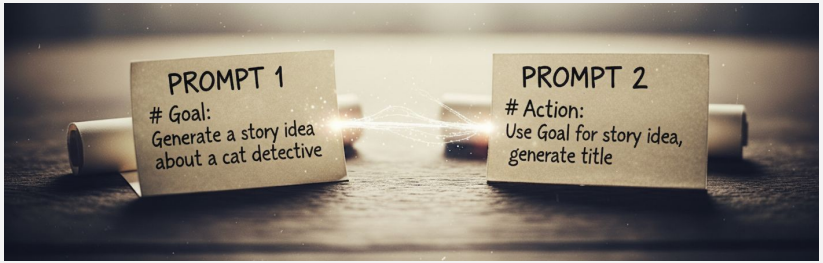
**Context-Packer + Actor**

- `pack.md` → condenses messy inputs (docs/links/notes) into **BRIEF | CITES[] | RISKS | OPEN_QS**
- `act.md` → executes strictly from BRIEF; must cite CITES; resolve OPEN_QS or return TODOs

**Evidence-Weaver + Synthesizer**

- `gather.md` → pulls multi-source facts with **QUOTES[], SOURCE[], LIMITS, TIMESTAMP**
- `weave.md` → produces claims; every claim maps to **QUOTES**[source]; unmet **LIMITS** become **QUESTIONS**

- Why this wins: tiny, auditable, swappable pieces → reliable outputs



# GOVERNANCE POWERED BY AGENTIC PAIRINGS

List your own agentic pairings of 1,2,3 or 4

**Once complete add both post your combination to the use-case-help channel**

### 1️⃣ Planner + Doer

- `planner.md` → turns a **GOAL** into ≤5 steps with inputs/outputs & tool hints
- `doer.md` → executes **step-by-step,** logs results, stops on rubric fail

### 2️⃣ Proposer + Critic

- `propose.md` → drafts full (copy/brief) & (plan/decision)
- `critic.md` → scores against a rubric; if <7/9, returns **one fix** then revise once

### 3️⃣ Context-Packer + Actor

- `pack.md` → condenses messy inputs (docs/links/notes) into **BRIEF | CITES[] | RISKS | OPEN_QS**
- `act.md` → executes strictly from BRIEF; must cite CITES; resolve OPEN_QS as TODOs

### 4️⃣ Choose Your Own Pairing

Create any two markdown files that work together adversarially or collaboratively to achieve goals

💥 Ask: “How can I solve a bigger problem with simple .md files?

# Time to work
# + Open Q&A

# WEEKLY CHECK-IN 5/20 RECAP

Agents are powered by these core capabilities:

**Autonomy**
How far the agent can run independently without human intervention.

- Choose Based on Context Delta

**Reasoning & Planning**
Turning high-level goals into clear, checkable execution steps.

- Build A Stateful Engine

**Memory**
Scratchpad → Episodic log → Reusable knowledge notes.

- Memory is metabolic not static

**Tool Use**
Using APIs and apps as verbs with structured inputs and outputs.

- Unify Tool Use in World Models

**Governance**
Defining the essential rules and safety guidelines for the agent.

- Leverage Agentic Pairings

# What's Next